



WRIGHT STATE UNIVERSITY

Welcome to Introduction to Databases & Modeling - Section 01

Fall 2026 CS 3700 01

MoWeFr 1010 AM 1105 AM | Main Russ Engineer Cntr 150

Instructor Information

Bibek Joshi
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Wright State University
MS in Computer Science

Instructor Information (hidden)

Office Location
Russ Engineering Center RC 331
Office Hours
Mondays, Tuesdays, Wednesdays, & Thursday 11:30 AM – 12:30 PM

Course Information

Relational client server database design and access techniques. Includes modeling and database design and writing SQL and PL/SQL statements. .

Credit Hours: 3

Prerequisite(s): CS 1180 CS 1160 CEG 2170

This course introduces the fundamental concepts of database systems with an emphasis on data modeling and relational design. Topics include the role and historical development of database management systems and logical data models, conceptual modeling using Entity-Relationship (ER) diagrams, relational schema design, SQL programming, normalization, transactions, integrity constraints, indexing, security, basic database administration, stored procedures and triggers, enterprise database systems, and an introduction to non-relational databases. Students will design, implement, administer, and query relational databases in the context of real-world and enterprise applications.

Course Objectives

To be able to model, create and maintain a relational database. To become proficient at SQL. To become proficient in using modern modeling software to model a relational database To understand the purpose and intricacies of an entity-relationship model and use it to design a relational database. To normalize a database. To learn critical-thinking techniques for solving unstructured problems by identifying and analyzing information systems problems and designing and implementing a working database system. To understand the basics of database administration.

Course Modality

Face-to-Face

Required Readings

NA

Recommended Reading

Database Design for Mere Mortals: A Hands-on Guide to Relational Database Design

ISBN: 0136788041

MySQL Crash Course: A Hands-on Introduction to Database Development

ISBN: 1718503008

Technology Requirements

To participate in learning activities and complete assignments, you will need:

- Access to a working computer that has a current operating system with updates installed.
- Reliable Internet access
- A WSU email account
- A current Internet browser that is compatible with Pilot such as Chrome
- Microsoft Office
- Reliable data storage for your work, such as a USB drive or Office365 OneDrive cloud storage

Grading Criteria

Labs / Homework: 30% Midterm: 20% Final Exam: 20% Project: 25% Participation: 5%

Grading Scale

- 90–100: A
- 80–89: B
- 70–79: C
- 60–69: D
- 0–59: F

Final grades are at the discretion of the instructor.

If you have personal concerns about grading, course progress, or extension requests, please email the course instructor directly.

Course Schedule

Week 1 - Introduction to Databases & DBMS Concepts

Week 2 - Conceptual Data Modeling I : ER fundamentals

Week 3 - Conceptual Data Modeling II : Advanced ER design

Week 4 - The Relational Data Model and ER-to-Relational Mapping

Week 5 - Schema Quality, Anomalies and Functional Dependencies

Week 6 - Normalization and Schema Refinement

Week 7 - Midterm Examination and Design Integration

Week 8 - SQL Data Definition Language (DDL): Schema Implementation

Week 9 - SQL Data Manipulation Language (DML): Data Management

Week 10 - SQL Queries I : Basic Retrieval

Week 11 - SQL Queries II: Joins and Aggregation

Week 12 - SQL Queries III: Subqueries and Advanced Queries

Week 13 - Transactions, Performance and Security

Week 14 - Modern Database Systems and NoSQL

Week 15 - Final Exam and Review

Student Attendance

Attendance is mandatory and will be recorded periodically. This is an in-person course, you are expected to attend all sessions. If you miss class due to illness or personal reasons, you are responsible for catching up on material and practice activities and providing early notice when possible.

Communication Guidelines

- Pilot for assignment submissions, course materials, and grades.
 - Pilot
- Discord will be the primary platform for class communication, questions, and troubleshooting.
 - Wright State CSE Discord Server: <https://discord.gg/pMUMzTt>
 - Course channels: #cs-class-selection . You will need to click on the selection before you can access the discord channel.
 - Verified accounts appear in green or gold, depending on role.

Note: It is the student's responsibility to check Pilot, Wright State email, and the course Discord server regularly (at least once daily) for announcements, resources, and clarifications.

Academic Integrity Policy

Students must adhere to Wright State University's academic integrity standards.

Academic dishonesty includes (but is not limited to):

- Cheating
- Plagiarism
- Misrepresentation of work
- Use of generative AI tools to create or edit assignments

All work submitted must be original. Any external sources must be properly cited.

Unauthorized collaboration or resubmission of prior work without explicit permission will be treated as plagiarism.

Sanctions may include:

- A zero on the assignment
- Reduction of the final grade
- Failure of the course

Important Dates

Access the important dates on the [Academic Calendar](#)

University Resources

- [Office of Disability Services](#)
- **Religious Accommodations (Dayton and Lake)**
To request an accommodation, follow the procedure in [University Policy 3380](#). Requests must be submitted in writing and during the first 14 days of the semester. Questions about this Policy may be directed to the Office of General Counsel at generalcounsel@wright.edu, or (937) 775-3540.
- [University Writing Center](#)
- [Math Learning Center](#)
- [Academic Advising](#)
- [Counseling and Wellness](#)
- [Student Resources and Support](#)
- [Wright State Bookstore \(Dayton and Lake\)](#)